

Homework 4

Due Date: 2:00 pm, May 25 (Monday)

Problem 1. For any linear programming problem in standard form and its dual problem, label each of the following statements as true or false and justify your answer.

- (a) The sum of the number of functional constraints and the number of variables, before augmenting, is the same for both the primal and the dual problems.
- (b) At each iteration, the simplex method simultaneously identifies a corner-point feasible solution for the primal problem and a corner-point feasible solution for the dual problem such that their objective function values are the same.
- (c) If the primal problem has an unbounded objective function, then the optimal value of the objective function for the dual problem must be zero.

Problem 2. Consider the following problem.

$$\begin{array}{ll}\text{Maximize} & Z = 2x_1 - 4x_2, \\ \text{subject to} & x_1 - x_2 \leq 1, \\ & x_1, x_2 \geq 0.\end{array}$$

- (a) Construct the dual problem, and then find its optimal solution by inspection.
- (b) Use the complementary slackness property and the optimal solution for the dual problem to find the optimal solution for the primal problem.
- (c) Suppose that c_1 , the coefficient of x_1 in the primal objective function, actually can have any value in the model. For what values of c_1 does the dual problem have no feasible solutions? For these values, what does duality theory then imply about the primal problem?

Problem 3. Consider the following primal-dual pair.

$$\begin{array}{ll}\text{Primal: Minimize} & 13x_1 + 10x_2 + 6x_3 \\ \text{subject to} & 5x_1 + x_2 + 3x_3 \geq 8, \\ & 3x_1 + 2x_2 \geq 3, \\ & x_1, x_2, x_3 \geq 0,\end{array}$$

$$\begin{array}{ll}\text{Dual: Maximize} & 8y_1 + 3y_2 \\ \text{subject to} & 5y_1 + 3y_2 \leq 13, \\ & y_1 + 2y_2 \leq 10, \\ & 3y_1 \leq 6, \\ & y_1, y_2 \geq 0.\end{array}$$

- (a) Write down the complementary slackness conditions.
- (b) If the primal optimal solution is given by $(x_1, x_2, x_3) = (1, 0, 1)$, use complementary slackness to find a dual optimal solution.

Problem 4. A bookmaker accepts orders for a football tournament. Exactly one of three teams, Red (R), Blue (B), and Green (G), will win. Each accepted unit of order j pays one dollar under the outcomes indicated in the table.

j	$a_j = (a_{Rj}, a_{Bj}, a_{Gj})^T$	π_j	q_j
1	$(1, 0, 0)^T$	$\frac{9}{20}$	2
2	$(0, 1, 0)^T$	$\frac{7}{20}$	2
3	$(0, 0, 1)^T$	$\frac{3}{10}$	2
4	$(1, 1, 0)^T$	$\frac{13}{20}$	1

Here π_j is the price collected per unit and q_j is the maximum quantity available. Let x_j be the accepted quantity of order j , and let w be the worst-case payoff of all accepted orders.

- (a) Write the bookmaker's primal linear programming model.
- (b) Derive the dual problem. Explain the meaning of the dual variables p_R, p_B, p_G and y_1, y_2, y_3, y_4 .
- (c) Using complementary slackness, state when an order is fully accepted, partially accepted, or not accepted.
- (d) Find one optimal primal-dual pair.